F78PA Probability & Statistics A

Tutorial 5 – Solutions

1. Suppose that $X \sim \mathrm{N}(4, 25)$. Standardise: $Z = \frac{X-4}{\sqrt{25}} \sim \mathrm{N}(0, 1)$.

   (a)

$$\begin{aligned}
\mathbb{P}(X > 6) &= \mathbb{P}\left(\frac{X-4}{\sqrt{25}} > \frac{6-4}{\sqrt{25}}\right) \\
&= \mathbb{P}(Z > 0.4) \qquad \text{where } Z \sim \mathrm{N}(0,1) \\
&= 1 - \mathbb{P}(Z \leq 0.4) \\
&= 1 - \Phi(0.4) \\
&= 1 - 0.6554 \qquad \text{from tables} \\
&= 0.3446
\end{aligned}$$

   (b)

$$\begin{aligned}
\mathbb{P}(X < 2) &= \mathbb{P}\left(\frac{X-4}{\sqrt{25}} < \frac{2-4}{\sqrt{25}}\right) \\
&= \mathbb{P}(Z < -0.4) \qquad \text{where } Z \sim \mathrm{N}(0,1) \\
&= \Phi(-0.4) \\
&= 1 - \Phi(0.4) \\
&= 1 - 0.6554 \qquad \text{from tables} \\
&= 0.3446
\end{aligned}$$

   (c)

$$\begin{aligned}
\mathbb{P}(0 < X < 10) &= \mathbb{P}\left(\frac{0-4}{\sqrt{25}} < \frac{X-4}{\sqrt{25}} < \frac{10-4}{\sqrt{25}}\right) \\
&= \mathbb{P}(-0.8 < Z < 1.2) \qquad \text{where } Z \sim \mathrm{N}(0,1) \\
&= \mathbb{P}(Z < 1.2) - \mathbb{P}(Z \leq -0.8) \\
&= \Phi(1.2) - \Phi(-0.8) \\
&= \Phi(1.2) - (1 - \Phi(0.8)) \\
&= 0.8849 - (1 - 0.7881) \qquad \text{from tables} \\
&= 0.673
\end{aligned}$$

2. We are told that $X \sim \mathrm{N}\left(5, \sigma^2\right)$, and required to determine the value of σ^2 such that $\mathbb{P}(X \geq 9) = 0.3$.

   Standardising, $Z = \frac{X-5}{\sigma} \sim \mathrm{N}(0, 1)$. Now

$$\begin{aligned}
\mathbb{P}(X \geq 9) &= \mathbb{P}\left(\frac{X-5}{\sigma} \geq \frac{9-5}{\sigma}\right) \\
&= \mathbb{P}\left(Z \geq \frac{4}{\sigma}\right) \text{ where } Z \sim \mathrm{N}(0,1)
\end{aligned}$$

   So we have $\mathbb{P}(Z \geq \frac{4}{\sigma}) = 0.3$, where $Z \sim \mathrm{N}(0, 1)$.

   From Table 5 (percentage points of the $\mathrm{N}(0, 1)$ distribution), we find

$$\mathbb{P}(Z \geq 0.5244) = 0.3$$

   .

   It follows that $\frac{4}{\sigma} = 0.5244$, and solving for σ we obtain $\sigma = 4/0.5244 \approx 7.628$.

   So $\mathbb{V}\mathrm{ar}(X) = \sigma^2 \approx 58.183$.

3. We are told that the time in minutes, T, that it takes a student to travel from home to university is a random variable distributed as $T \sim \mathrm{N}\left(40, 7^2\right)$.

   Let t be the number of minutes before the lecture start time (9:20am) that the student leaves home.

   The student will be on time provided $T \leq t$.

   We need to find t such that $\mathbb{P}(T \leq t) = \mathbb{P}(\text{Student is on time}) \geq 0.95$.

   Standardising, $Z = \frac{T-40}{7} \sim \mathrm{N}(0,1)$.

   Now

$$\begin{aligned}
\mathbb{P}\left(\text{Student is on time}\right) &= \mathbb{P}(T \leq t) \\
&= \mathbb{P}\left(\frac{T-40}{7} \leq \frac{t-40}{7}\right) \\
&= \mathbb{P}\left(Z \leq \frac{t-40}{7}\right) \qquad \text{where } Z \sim \mathrm{N}(0,1)
\end{aligned}$$

   We need to find the value of t such that $\mathbb{P}\left(Z \leq \frac{t-40}{7}\right) = 0.95$.

   From Table 5 (percentage points of the $\mathrm{N}(0,1)$ distribution), we find $\mathbb{P}(Z \leq 1.6449) = 0.95$.

   So we must have $\frac{t-40}{7} = 1.6449$, and solving for t we find $t = 7 \times 1.6449 + 40 = 51.51$.

   So the student must leave home 52 minutes (approximately) before the lecture start time of 9:20am.

   That is, the student must leave home by 8:28am.

4. (a) $Y = \dfrac{1}{\sqrt{X+3}}$

   Since $X \sim U(0,1)$, then $0 \leq X \leq 1$, so $3 \leq X+3 \leq 4$, so $\sqrt{3} \leq \sqrt{X+3} \leq 2$, and so $\frac{1}{2} \leq \frac{1}{\sqrt{X+3}} \leq \frac{1}{\sqrt{3}}$. That is, $\mathrm{Im}Y = \left[\frac{1}{2}, \frac{1}{\sqrt{3}}\right]$.

   Since Im $Y = \left[\frac{1}{2}, \frac{1}{\sqrt{3}}\right]$, then $F_Y(y) = 0$ for $y < 1/2$, and $F_Y(y) = 1$ for $y > 1/\sqrt{3}$. For $\frac{1}{2} \leq y \leq \frac{1}{\sqrt{3}}$,

$$\begin{aligned}
F_Y(y) = \mathbb{P}(Y \leq y) &= \mathbb{P}\left(\frac{1}{\sqrt{X+3}} \leq y\right) \\
&= \mathbb{P}\left(\sqrt{X+3} \geq \frac{1}{y}\right) \\
&= \mathbb{P}\left(X+3 \geq \frac{1}{y^2}\right) \\
&= \mathbb{P}\left(X \geq \frac{1}{y^2} - 3\right) \\
&= \mathbb{P}\left(\frac{1}{y^2} - 3 \leq X \leq 1\right) \quad (\text{since } \mathbb{P}(X > 1) = 0)
\end{aligned}$$

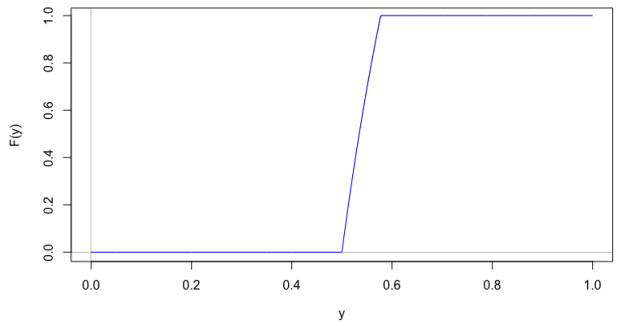
   Since X is uniformly distributed on $[0,1]$,

$$\begin{aligned}
\mathbb{P}\left(\frac{1}{y^2} - 3 \leq X \leq 1\right) &= \frac{\text{Length of interval } \left[\frac{1}{y^2} - 3, 1\right]}{\text{Length of interval } [0,1]} \\
&= \frac{1 - \left(\frac{1}{y^2} - 3\right)}{1} = 4 - \frac{1}{y^2}
\end{aligned}$$

So we have

$$F_Y(y) = \begin{cases} 0 & y < 1/2, \\ 4 - \frac{1}{y^2} & 1/2 \le y \le 1/\sqrt{3}, \\ 1 & y > 1/\sqrt{3}. \end{cases}$$

Sketch of $F_Y(y)$:



Since $F_Y(y)$ is continuous and differentiable almost everywhere, Y has a density function, given by

$$f_Y(y) \;=\; F_Y'(y) \;=\; \begin{cases} \frac{2}{y^3} & 1/2 \le y \le 1/\sqrt{3}, \\ 0 & \text{otherwise.} \end{cases}$$

(b) $W = \max\{1, -\log X\}$

Since $X \sim \mathrm{U}(0,1)$, we have $0 \le X \le 1$, so $0 \le -\log X < \infty$, and so $1 \le \max\{1, -\log X\} < \infty$. That is, Im $W = [1, \infty)$.

Since Im $W = [1, \infty)$, then $F_W(w) = 0$ for $w < 1$. For $w \ge 1$,

$$\begin{aligned} F_W(w) \;=\; \mathbb{P}(W \le w) \;&=\; \mathbb{P}\left(\max\{1, -\log X\} \le w\right) \\ &=\; \mathbb{P}\left(-\log X \le w\right) \quad \text{(because } w \ge 1 \text{ here)} \\ &=\; \mathbb{P}\left(\log X \ge -w\right) \\ &=\; \mathbb{P}(X \ge e^{-w}) \\ &=\; \mathbb{P}(e^{-w} \le X \le 1) \quad \text{(since } \mathbb{P}(X > 1) = 0) \end{aligned}$$

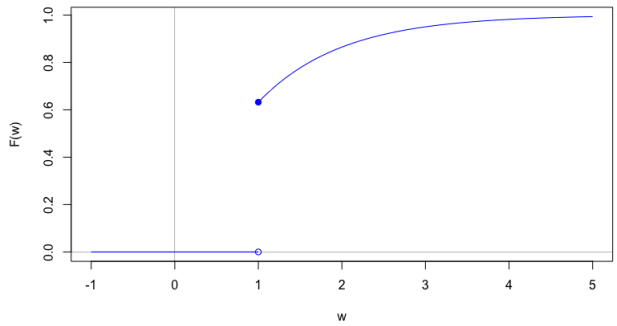
Since X is uniformly distributed on $[0,1]$,

$$\mathbb{P}(e^{-w} \le X \le 1) \;=\; \frac{\text{Length of interval } [e^{-w}, 1]}{\text{Length of interval } [0,1]} \;=\; \frac{1 - e^{-w}}{1} \;=\; 1 - e^{-w}$$

So we have

$$F_W(w) = \begin{cases} 0 & w < 1, \\ 1 - e^{-w} & w \ge 1. \end{cases}$$

Sketch of $F_W(w)$:



F_W is differentiable everywhere except at $w = 1$, but it is **not continuous**, and therefore W does **not** have a density.

(c) $V = -\log(X)$

Since $X \sim U(0,1)$, we have $0 \leq X \leq 1$, so $0 \leq -\log X < \infty$. That is, Im $V = [0, \infty)$.

Since Im $V = [0, \infty)$, then $F_V(v) = 0$ for $v < 0$. For $v \geq 0$,

$$\begin{aligned} F_V(v) &= \mathbb{P}(V \leq v) &&= \mathbb{P}(-\log X \leq v) \\ &&&= \mathbb{P}(\log X \geq -v) \\ &&&= \mathbb{P}(X \geq e^{-v}) \\ &&&= \mathbb{P}(e^{-v} \leq X \leq 1) \quad (\text{since } \mathbb{P}(X > 1) = 0) \end{aligned}$$

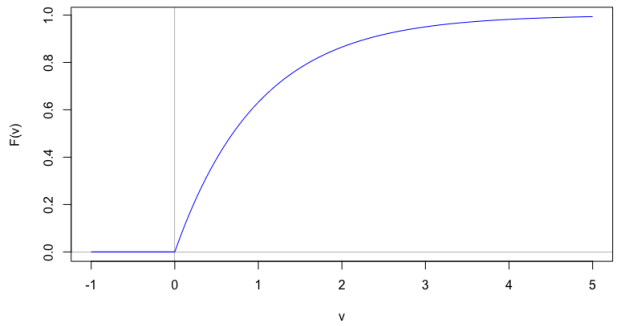
Since X is uniformly distributed on $[0, 1]$,

$$\mathbb{P}(e^{-v} \leq X \leq 1) = \frac{\text{Length of interval } [e^{-v}, 1]}{\text{Length of interval } [0, 1]} = \frac{1 - e^{-v}}{1} = 1 - e^{-v}$$

So we have

$$F_V(v) = \begin{cases} 0 & v < 0, \\ 1 - e^{-v} & v \geq 0. \end{cases}$$

Sketch of $F_V(v)$:



Since $F_V(v)$ is continuous and differentiable almost everywhere, V has a density function, given by

$$f_V(v) = F_V'(v) = \begin{cases} 0 & v < 0, \\ e^{-v} & v \geq 0. \end{cases}$$

5. (a) Example of a distribution function that is not continuous: many examples possible, for instance $F_X(x) = 1$ for $x \geq 0$ and $F_X(x) = 0$ for $x < 0$, this is the distribution function for the random variable $X = 0$.

   (b) Example of a function that takes values between 0 and 1 but which is not a distribution function: many examples possible, for instance $F(x) = 0$ for all x.

6. Denote by X the distance (in miles) from the breakdown to city A has a uniform distribution, then $X \sim \text{U}(0, 120)$.

   Let $Y = g(X)$ denote the distance to the closest service station from X, i.e.

$$g(x) = \min(x, |60 - x|, 120 - x).$$

   This function is plotted in Figure 1. The expected distance to the nearest service

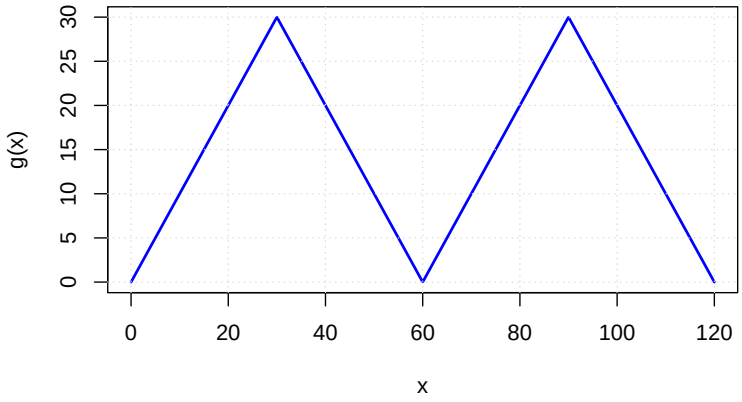


Figure 1: Plot of $g(x) = \min(x, |60 - x|, 120 - x)$ over the interval $[0, 120]$

station is $\mathbb{E}[g(X)]$ and is given by

$$\mathbb{E}[g(X)] = \frac{1}{120} \int_0^{120} g(x)\, \mathrm{d}x.$$

Note that g defines the outline of two triangles so we can calculate the area under the curve of g using the formula for the area of a triangle, i.e. $0.5 \times$ base $\times$ height. In this case both triangles have base of 60 and height of 30 so each triangle has an area of 900, therefore,

$$\int_0^{120} g(x)\, \mathrm{d}x = 2 \times 900 = 1800.$$

Hence,

$$\mathbb{E}[g(X)] = \frac{1}{120}(1800) = 15.$$

7. (a) Let $Y = aX + b$, where $a > 0$. Then for any $y \in \mathbb{R}$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \le y) &&= \mathbb{P}(aX + b \le y) \\ &&&= \mathbb{P}\left(X \le \frac{y-b}{a}\right) \quad \text{(since } a > 0\text{)} \\ &&&= F_X\left(\frac{y-b}{a}\right) \end{aligned}$$

Now differentiating (using the chain rule), we obtain

$$f_Y(y) = F_Y'(y) = \frac{d}{dy} F_X\left(\frac{y-b}{a}\right) = \frac{1}{a} F_X'\left(\frac{y-b}{a}\right) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

(b) Now suppose that $Y = aX + b$ but $a < 0$. The difference from the above argument is that now

$$\mathbb{P}(aX + b \le y) = \mathbb{P}\left(X \ge \frac{y-b}{a}\right) \quad \text{(since } a < 0\text{)}$$

Following through the algebra, we obtain

$$F_Y(y) = 1 - F_X\left(\frac{y-b}{a}\right)$$

and hence

$$f_Y(y) = -\frac{1}{a} f_X\left(\frac{y-b}{a}\right)$$

(c) Let X denote the distance, in centimetres, from the centre of the target. Then we know from Tutorial 4 Q6 that $\mathrm{Im}(X) = [0, 100]$ and

$$f_X(x) = \begin{cases} \frac{x}{5000} & 0 \le x \le 100, \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = 5 - \frac{X}{20}$. Then $\mathrm{Im}(Y) = [0, 5]$, and to find f_Y we apply part (b) above with $a = -1/20$, $b = 5$.
Since $\mathrm{Im}(Y) = [0, 5]$ we know that $f_Y(y) = 0$ for $y < 0$ and for $y > 5$.
For $0 \le y \le 5$, applying part (b),

$$f_Y(y) = -\frac{1}{-(1/20)}\ f_X\left(\frac{y-5}{-(1/20)}\right) = 20 f_X(100 - 20y) = 20\left(\frac{100 - 20y}{5000}\right) = \frac{2}{5} - \frac{2y}{25}$$

So

$$f_Y(y) = \begin{cases} \frac{2}{5} - \frac{2y}{25} & 0 \le y \le 5, \\ 0 & \text{otherwise.} \end{cases}$$

8. Suppose $Z \sim \mathrm{N}(0, 1)$.

Then denoting by $\varphi(z)$ the standard normal probability density function,

$$\mathbb{E}[Z] = \int_{z=-\infty}^{\infty} z\varphi(z)\, dz = \frac{1}{\sqrt{2\pi}}\left(\int_{z=-\infty}^{0} z e^{-z^2/2}\, dz + \int_{z=0}^{\infty} z e^{-z^2/2}\, dz\right)$$

Now making the substitution $s = z^2/2$, so that $ds = z\, dz$,

$$\int_{z=0}^{\infty} z e^{-z^2/2}\, dz \;=\; \int_{s=0}^{\infty} z e^{-s}\, \frac{ds}{z} = \int_{s=0}^{\infty} e^{-s}\, ds = \left[-e^{-s}\right]_{s=0}^{\infty} = -\lim_{M\to\infty} e^{-M} + 1 = 1$$

since $\lim_{M\to\infty} e^{-M} = 0$.
Using the same substitution $s = z^2/2$,

$$\int_{z=-\infty}^{0} z e^{-z^2/2}\, dz \;=\; \int_{s=+\infty}^{0} z e^{-s}\, \frac{ds}{z} = \int_{s=\infty}^{0} e^{-s}\, ds = \left[-e^{-s}\right]_{s=\infty}^{0} = -1 + \lim_{M\to\infty} e^{-M} = -1$$

So

$$\mathbb{E}[Z] = \frac{1}{\sqrt{2\pi}}(-1 + 1) = 0$$

as required.
To find $\mathbb{V}\mathrm{ar}[Z]$, first need to compute $E[Z^2]$.

$$E[Z^2] = \int_{z=-\infty}^{\infty} z^2 \varphi(z)\, dz \;=\; \int_{z=-\infty}^{\infty} z^2\, \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \frac{1}{\sqrt{2\pi}} \int_{z=-\infty}^{\infty} z^2\, e^{-z^2/2}\, dz$$

Now integrate by parts: setting

$$u = z, \qquad \frac{dv}{dz} = z e^{-z^2/2},$$

then we have

$$\frac{du}{dz} = 1, \qquad v = -e^{-z^2/2}.$$

[**Note:** this requires us to recognise $z e^{-z^2/2}$ as the derivative of $-e^{-z^2/2}$.]

Applying integration by parts formula,

$$\begin{aligned}\mathbb{E}[Z^2] &= \frac{1}{\sqrt{2\pi}}\left(\left[-ze^{-z^2/2}\right]_{z=-\infty}^{\infty} + \int_{z=-\infty}^{\infty} e^{-z^2/2}\,dz\right)\\ &= 0 + \int_{z=-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}\,dz\\ &= \int_{z=-\infty}^{\infty}\varphi(z)\,dz = 1 \qquad (\varphi(z) \text{ is a pdf, so integrates to 1}).\end{aligned}$$

Note: we have made use of the fact that $\lim_{z\to-\infty} ze^{-z^2/2} = \lim_{z\to+\infty} ze^{-z^2/2} = 0$.

Finally, we have $\mathbb{V}\mathrm{ar}[Z] = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 = 1 - 0^2 = 1$, as required.

9. Suppose that $Z \sim \mathrm{N}(0,1)$ and let $Y = Z^2$. Since $\mathrm{Im}\, Z = (-\infty, \infty)$, we have $\mathrm{Im}\, Y = [0, \infty)$.

   To show that $Y \sim \Gamma(\frac{1}{2}, \frac{1}{2})$, we find the density function $f_Y(y)$.

   Since $\mathrm{Im}Y = [0, \infty)$, we have $f_Y(y) = 0$ for $y < 0$.

   For $y \geq 0$, first consider the distribution function $F_Y(y)$.

$$\begin{aligned}F_Y(y) = \mathbb{P}(Y \leq y) &= \mathbb{P}(Z^2 \leq y)\\ &= \mathbb{P}\left(-\sqrt{y} \leq Z \leq \sqrt{y}\right)\\ &= \mathbb{P}\left(-\sqrt{y} < Z \leq \sqrt{y}\right) \qquad (\text{since } Z \text{ is a continuous r.v.})\\ &= \mathbb{P}(Z \leq \sqrt{y}) - \mathbb{P}(Z \leq -\sqrt{y})\\ &= \Phi(\sqrt{y}) - \Phi(-\sqrt{y})\end{aligned}$$

   where Φ is the distribution function of $\mathrm{N}(0,1)$.

   Differentiating, using the chain rule, we obtain

$$f_Y(y) = F_Y'(y) = \Phi'(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + \Phi'(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}}$$

   Now $\Phi'(x) = \varphi(x) = \dfrac{1}{\sqrt{2\pi}}e^{-x^2/2}$, the probability density function of $\mathrm{N}(0,1)$. So

$$\begin{aligned}f_Y(y) = \frac{e^{-y/2}}{2\sqrt{y}\,\sqrt{2\pi}} + \frac{e^{-y/2}}{2\sqrt{y}\,\sqrt{2\pi}} &= \frac{e^{-y/2}}{\sqrt{2\pi y}}\\ &= \frac{(\frac{1}{2})^{1/2} y^{-1/2} e^{-y/2}}{\Gamma(1/2)}\end{aligned}$$

   where we have re-arranged our formula to match the standard form of the Gamma probability density function, and made use of the hint that $\Gamma(1/2) = \sqrt{\pi}$.

   We have now shown that $Y = Z^2$ has density function

$$f_Y(y) - \begin{cases}\dfrac{(\frac{1}{2})^{1/2} y^{-1/2} e^{-y/2}}{\Gamma(1/2)} & y \geq 0,\\ 0 & \text{otherwise.}\end{cases}$$

   This is the density for a $\Gamma(\frac{1}{2}, \frac{1}{2})$ random variable, and it follows that $Y \sim \Gamma(\frac{1}{2}, \frac{1}{2})$, as required.

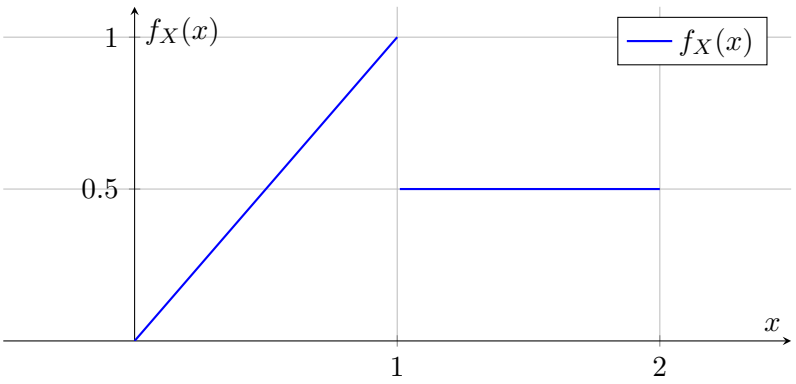


Figure 2: Plot of the piecewise function $f_X(x)$ defined as x for $0 \leq x \leq 1$, $\frac{1}{2}$ for $1 < x \leq 2$, and 0 otherwise.

Exam Style Question

(a) The function f_X is sketched in Figure 2. [1 mark].

To verify that $f_X(x)$ is a valid probability density function we check the two required conditions:

- **Non-negativity:**
  The function $f_X(x)$ is non-negative for all $x \in \mathbb{R}$, since it is defined as x on $[0, 1]$, $\frac{1}{2}$ on $(1, 2]$, and 0 elsewhere.
- **Total area under the curve equals 1:**
  We compute the integral of $f_X(x)$ over its support:

$$\int_{-\infty}^{\infty} f_X(x)\, \mathrm{d}x = \int_0^1 x\, \mathrm{d}x + \int_1^2 \frac{1}{2}\, \mathrm{d}x = \left[\frac{x^2}{2}\right]_0^1 + \left[\frac{x}{2}\right]_1^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

[1 mark]

(b) The image of X is determined by the set of values where f_X is non-negative so $\mathrm{Im}(X) = [0, 2]$. [1 mark]

(c) The expected value is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, \mathrm{d}x = \int_0^1 x \cdot x\, \mathrm{d}x + \int_1^2 x \cdot \frac{1}{2}\, \mathrm{d}x = \left[\frac{x^3}{3}\right]_{x=0}^1 + \left[\frac{x^2}{4}\right]_{x=1}^2 = \frac{1}{3} + 1 - \frac{1}{4} = \frac{13}{12}.$$

[2 marks]

(d) Since $\mathrm{Im}(X) = [0, 2]$ we have $F_X(x) = 0$ for $x \leq 0$ and $F_X(x) = 1$ for $x \geq 2$. [1 mark]

For $0 < x < 1$,

$$F_X(x) = \int_0^x f_X(u)\, \mathrm{d}u = \int_0^x u\, \mathrm{d}u = \frac{x^2}{2}.$$

For $1 \leq x < 2$

$$\begin{aligned} F_X(x) &= F_X(1) + \int_1^x f_X(u)\, \mathrm{d}u \\ &= \frac{1}{2} + \int_1^x \frac{1}{2}\, \mathrm{d}u \\ &= \frac{1}{2} + \frac{1}{2}(x - 1) = \frac{x}{2}. \end{aligned}$$

To summarise

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{x^2}{2}, & 0 < x < 1, \\ \frac{x}{2}, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

[2 marks]

(e) i) Since $h(x) = e^x$ is a strictly increasing function we know that $0 \leq x \leq 2$ if and only if $e^0 \leq e^x \leq e^2$ so $\mathrm{Im}(Y) = [1, e^2]$. [1 mark]
For $1 \leq y \leq e^2$ we have

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) && \text{by definition of } F_Y \\ &= \mathbb{P}(e^X \leq y) && \text{by definition of } Y \\ &= \mathbb{P}(\log(e^X) \leq \log(y)) && \text{since log is strictly increasing} \\ &= \mathbb{P}(X \leq \log(y)) = F_X(\log(y)). \end{aligned}$$

[2 marks]

ii) For $y \notin [1, e^2]$ we have $f_Y(y) = 0$.
[1 mark]
For $y \in [1, e^2]$ we can find $f_Y(y)$ by differentiating $F_Y(y)$ (this is valid for $0 < y < e$ and $e < y < e^2$ since F_Y is smooth for these values). By part d)i) we have

$$F_Y(y) = F_X(\log(y))$$

so by the chain rule

$$f_Y(y) = F_Y'(y) = F_X'(\log(y))\frac{d}{dy}(\log(y)) = \frac{f_X(\log(y))}{y}.$$

[1 mark]

Therefore

$$f_Y(y) = \begin{cases} \frac{\log(y)}{y}, & \text{if } 0 \leq \log(y) \leq 1, \\ \frac{1}{2y}, & \text{if } 1 < \log(y) \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Rearranging the constraints gives

$$f_Y(y) = \begin{cases} \frac{\log(y)}{y}, & \text{if } 1 \leq y \leq e, \\ \frac{1}{2y}, & \text{if } e < y \leq e^2, \\ 0, & \text{otherwise.} \end{cases}$$

[1 mark]